



Substitution of cane molasses for corn grain at two levels of degradable protein. I. Lactating cow performance, nutrition model predictions, and potential basis for butterfat and intake responses

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ABSTRACT

Little data is presently available on our ability to predict the combined effect of modifying diets with feeds rich in sugars or starch (ST) and rumen-degradable protein (RDP) on the performance of high-producing dairy cows. The objective of this study was to compare responses of 59 lactating Holstein cows to substitution of cane molasses (Mol) for dry corn grain (CG) at 3 levels of Mol and 2 levels of RDP (+RDP or –RDP) in a randomized complete block design with a 3×2 factorial arrangement of treatments. Also, lactation responses predicted by 2 nutritional models were compared with observed responses, with Mol composition entered so that nonnutritive materials in Mol were not counted as potentially digestible carbohydrate. We hypothesized that dry matter (DM) intake and milk fat percentage responses would increase with increasing Mol and would potentially be greater with +RDP. For evaluation of the nutritional models, we adopted the null hypothesis that observed and predicted lactation performance would not differ. Cows were individually fed a common diet during a 2-wk covariate period followed by 8 wk on experimental diets. Diets were formulated to be isonitrogenous and provide similar amounts of ST and water-soluble carbohydrates. Experimental diets contained, on a DM basis, 35% corn silage, 20% alfalfa silage, and 16.6% crude protein. The 0, 5.25, and 10.5% Mol diets respectively contained 19.0, 14.5, and 10.0% CG; 28, 25, and 22% ST; and 5.5, 8.5, and 11.5% water-soluble carbohydrates. At 10 wk on study, cows averaged 45.5 kg of energy-corrected milk (ECM). The DM intake (DMI), and yields of milk, milk protein, and ECM, and milk N/intake N declined linearly with increasing Mol. Differences among diets were not detected for milk fat yield and ECM/DMI. No RDP

or interaction effects were detected for these measures. That milk production efficiency did not differ across diets suggests that DMI was a primary driver of performance. The similar ECM/DMI and maintenance of milk fat yield would not have been predicted based on Mol and CG composition but may relate to differences in fermentation rates and products. As explanation for these results, we hypothesize that more rapid ruminal evolution of volatile fatty acids post-ingestion with Mol compared with CG may have provided masses of acetate and butyrate in excess of existing energy and synthetic needs that were shunted to milk fat production, and of propionate that depressed intake. The 2001 Dairy National Research Council model and the Cornell Net Carbohydrate and Protein System 6.55 in Nutritional Dynamic System Professional (2021) estimates of metabolizable protein-allowable ECM underestimated actual ECM for +RDP diets by 4.5 and 2.3 kg, respectively, and came close or overestimated for –RDP diets by 0.25 and 5.0 kg, respectively. Prediction discrepancies suggest issues with valuation of dietary protein based on degradability. Improved understanding of factors mediating these results would likely enhance our ability to predict animal responses.

Key words: molasses, rumen-degradable protein, dairy cow, lactation performance

INTRODUCTION

Sucrose and starch are NFC that differ from each other in ruminal fermentation characteristics. Sucrose disappears rapidly in the rumen (up to 738%/h; Weisbjerg et al., 1998), whereas starch ferments more slowly. Compared with starch, fermentation of sucrose by ruminal microbes produces more butyrate (Strobel and Russell, 1986), which is an important constituent of de novo synthesized fatty acids (FA) in milk (Palmquist et al., 1969). A commonly noted difference between these NFC is that providing more sucrose and decreasing starch in lactating cow diets can increase milk fat production (Broderick et al., 2008), but responses vary.

Received September 2, 2021.

Accepted January 10, 2022.

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Rumen-degradable protein supply may play a role in determining responses to carbohydrates, but few studies have investigated the interaction. Increasing the amount of RDP available to ruminal microbes at a given amount of NFC has been shown to increase microbial yield *in vitro* (Argyle and Baldwin, 1989; Stokes et al., 1991). Increasing RDP also increases the flux of carbohydrate through glycolysis (Malestein et al., 1984), which indicates an increased rate of fermentation of the carbohydrate. Variation in RDP supply has been shown to affect the impact of NFC on NDF digestibility *in vivo*. Relative to controls, supplementation with sugars increased total-tract NDF digestibility in steers when more RDP was supplemented but reduced it when less was provided (Heldt et al., 1999).

The influence of the NFC and RDP interaction on animal performance seems to contradict responses expected based on *in vitro* data. Efficiency of fat- and protein-corrected milk production showed an RDP $\times$ carbohydrate source interaction when corn grain, citrus pulp, and sucrose plus cane molasses were the supplements, decreasing with more RDP in all but the corn grain diet (Hall et al., 2010). In another study, DMI was the sole performance measure that tended to show a carbohydrate $\times$ RDP interaction, with DMI increasing with increasing glucose substitution for starch and rising more with the diets lower in RDP (diets provided up to 10% glucose and contained 61% forage; Sun et al., 2019).

A challenge faced in diet formulation to test the effects of dietary RDP is that RDP is not a single value, although relative differences between diets may be achieved through selection of protein sources. Ruminally degradable protein is a function of the competing rate of degradation of the protein source by rumen microbes and its rate of passage from the rumen (NRC, 2001). These have potential to be influenced by several factors, including level of DMI and diet composition, and the influence of forage and fiber from forage on ruminal retention. Ruminal ammonia nitrogen greater than 50 mg/L (3.6 mM ammonia) has been offered as an indicator that RDP is adequate to avoid deficiency issues such as depressed fiber digestibility (Satter and Slyter, 1974).

Although nutritional models can give us a framework to test our understanding of dietary factors and their effects on animal response, it is rare to find published literature in which cows were offered diets varying in dietary sugar and starch contents specifically for evaluation of lactation performance predictions of nutritional models. The study of Higgs et al. (2013) assessed provision of starch-, fiber-, or sugar-based supplements to animals on pasture and did not detect differences

between the Cornell Net Carbohydrate and Protein System (CNCPS) predictions and actual milk production, but milk production was low (~ 25.4 kg of milk/d).

Few studies have explored the effects of changing dietary RDP on animal performance with different NFC supplied from feeds rather than from purified sources. Cane molasses and corn grain are common, commercially available feed sources of sugars and starch, respectively. The objectives of this study were to compare performance responses of lactating cows to 3 levels of cane molasses substituted for ground dry corn grain at 2 levels of RDP, and to evaluate the accuracy of nutritional model predictions of animal performance as described with Nutritional Dynamic System (NDS) Professional/CNCPS 6.55 (2021; RUM&N Sas) and the Dairy NRC model (NRC, 2001). Because the focus of the model evaluation was to assess the accuracy of the models, not the effects of known errors in feedstuff descriptions, the nonnutritive materials in cane molasses that do not analyze by the usual feed analysis methods were allocated to fractions that would not be used to produce energy or ruminal microbes. We hypothesized that DMI and milk fat percentage responses would increase with increasing levels of molasses, and would potentially be greater with higher levels of RDP. For evaluation of the nutritional models, we adopted the null hypothesis that observed and predicted lactation performance would not differ.

MATERIALS AND METHODS

The study was performed as a randomized complete block design with a 3×2 factorial arrangement of dietary treatments and cow as the experimental unit. Lactating dairy cows were individually offered a common diet during a 2-wk covariate period, and then offered experimental diets for 8 wk, for a total of 10 wk on experiment. Dietary treatments provided linear substitution of dry ground corn grain (CG) and cane molasses (Mol) at 19.0, 14.5, and 10.0%, and at 0, 5.25, and 10.5% of diet DM, respectively (Table 1). Carbohydrate source treatments were imposed at 2 levels of RDP (+RDP or -RDP), achieved through feeding soybean meal or expeller soybean meal (SoyPlus, Landus Cooperative), the latter providing a greater proportion of heat-treated protein with reduced RDP. Experimental diets were designed to be isonitrogenous and contain equal proportions of forage, starch + water-soluble carbohydrates + total sugars as invert, and calcium, and similar nitrogen-to-sulfur ratios. Compositions of Mol, CG, and forages used in the experimental diets are shown in Table 2.

A total of 59 noncannulated lactating Holstein cows, averaging (mean $\pm$ SD) 101 ± 22.8 DIM, 655 ± 88.7 kg BW, 1.8 ± 0.9 parity, and 49.0 ± 7.7 kg of milk/d at the beginning of this study, were maintained under protocols approved by the University of Wisconsin Institutional Animal Care and Use Committee (Madison, WI). We planned for 60 cows to be used, but 1 was lost to injury not related to the study; therefore $n = 10$ for

all treatments except 10.5% Mol +RDP, where $n = 9$. Cows were housed in individual tiestalls bedded with rubber mattresses and chopped straw at the USDA-ARS US Dairy Forage Research Center Farm in Prairie du Sac, Wisconsin. All cows were milked 3 times per day at approximately 0400, 1100, and 1830 h, and fed individually at approximately 0800 h once per day to a 5% refusal. Within lactation number (parities of 1, 2,

Table 1. Feed and chemical composition and particle size of diets during sampling weeks

Item	Cov ¹	+RDP ²			–RDP ²		
		Molasses, % of DM			Molasses, % of DM		
		0	5.25	10.5	0	5.25	10.5
Feed ingredients, % of DM							
Corn silage, brown midrib	28.6	35.5	35.4	35.2	35.6	35.5	35.4
Alfalfa silage	30.6	19.9	19.9	19.8	20.0	20.0	19.9
Distillers grains	2.31	2.0	2.0	2.0	2.0	2.0	2.0
High-moisture corn	20.8						
Canola meal	7.38						
Roasted soybeans	7.84						
Cane molasses		0	5.24	10.63	0	5.25	10.6
Dry ground corn		18.2	13.9	9.6	18.2	14.0	9.7
Soybean meal		13.5	13.6	13.9	6.6	6.6	6.7
SoyPlus ³		0	0	0	7.5	7.7	7.9
Soyhulls		7.8	7.1	6.3	6.8	6.0	5.4
KMgS minerals		0.28	0.15	0	0.28	0.15	0
Limestone		0.28	0.15	0	0.28	0.15	0
Vitamin-mineral mix ⁴	2.50	2.6	2.6	2.6	2.6	2.6	2.6
Chemical composition ⁵							
DM, %	49.4	53.2	52.8	52.4	53.1	52.7	52.3
Percentage of DM							
CP	16.4	16.7	16.7	16.7	16.6	16.5	16.5
aNDFom	27.3	26.9	26.2	25.3	27.0	26.2	25.4
Starch	24.9	27.9	24.8	21.6	28.3	24.9	21.7
WSC+TSI	3.9	5.5	8.5	11.7	5.5	8.5	11.6
Starch+WSC+TSI	28.8	33.5	33.3	33.3	33.5	33.5	33.3
Ash	8.1	8.27	8.52	8.74	8.25	8.50	8.71
Ether extract	3.3	4.1	3.9	3.7	4.5	4.3	4.1
Nitrogen/sulfur ratio	12.8	9.8	9.1	8.7	9.8	9.1	8.6
Forage NDF	22.9	19.2	19.0	19.0	19.2	19.1	19.0
Cumulative proportion of diet retained on sieve							
6.7 mm	1.8%	2.1%	2.6%	2.5%	4.3%	2.7%	3.2%
4.75 mm	9.3%	8.6%	7.6%	7.8%	9.8%	8.4%	9.3%
3.35 mm	22.5%	18.0%	15.5%	16.0%	20.8%	16.2%	17.7%
2.36 mm	37.2%	28.3%	24.0%	25.1%	31.1%	25.6%	28.2%
1.7 mm	49.7%	42.1%	37.2%	38.4%	44.8%	39.2%	41.7%
1.18 mm	56.2%	51.4%	46.1%	47.2%	53.7%	47.6%	50.3%
0.85 mm	68.8%	70.6%	64.7%	64.9%	70.8%	64.9%	66.4%
0.60 mm	78.5%	82.2%	77.0%	76.4%	82.0%	76.8%	76.7%
0.30 mm	91.0%	93.7%	90.6%	89.0%	93.6%	90.1%	88.4%
Pan	100.5%	100.1%	99.9%	100.1%	100.1%	99.9%	100.1%
Mean particle size, mm	1.45	1.37	1.19	1.19	1.46	1.33	1.26
Mean particle size, SD	0.48	0.42	0.45	0.47	0.45	0.38	0.49

¹Cov = covariate diet.

²RDP treatments: +, with more RDP; –, with less RDP.

³Expeller soybean meal; Landus Cooperative.

⁴Composition, % of DM: Ca 16.8, Mg 4.64, K 0.57, S 1.01, Na 15.8, Cl 7.12; as mg/kg: Co 45.4, Cu 553.6, I 64.2, Fe 778.3, Mn 2,776.2, Se 15.6 mg/kg, Zn 2,996.8. Vitamins, IU/kg: A 311,728, D₃ 62,390, E 1,453; monensin 523 mg/kg.

⁵aNDFom = NDF on an ash-free basis; TSI = total sugars as invert; WSC = water-soluble carbohydrates.

Table 2. Chemical composition (% of DM unless otherwise noted) and particle size of dry corn grain, cane molasses, and silages from final sampling week of study

Item	Corn grain	Cane molasses	Alfalfa silage	Corn silage	Soybean meal	Expeller soybean meal
Composition ¹						
DM, %	84.6	67.1	45.3	38.0	90.2	90.1
CP	8.22	7.9	21.3	6.8	52.2	46.6
NDF CP	1.02	—	3.11	1.74	0.79	9.02
ADF CP	0.45	—	1.45	0.81	0.68	0.92
aNDFom	6.41	—	34.8	34.3	6.93	15.8
Starch	71.9	—	2.7	39.2	1.51	1.51
WSC+TSI	3.55	60.4	3.0	3.4	18.2	16.6
Ether extract	4.01	<0.2	4.1	5.2	2.48	6.48
Ash	1.20	17.0	13.3	2.7	7.57	6.86
Unaccounted	4.70	14.8	20.8	8.4	11.1	6.15
Proportion of diet retained on sieve, %						
13.2 mm			0.89	1.86		
9.5 mm			1.17	3.35		
6.7 mm			2.87	10.38		
4.75 mm			7.14	19.10	0.13	0.10
3.35 mm	0.03		16.80	20.93	0.08	2.13
2.36 mm	7.58		19.16	13.88	1.51	7.78
1.70 mm	19.1		13.38	9.40	11.24	13.08
1.18 mm	14.6		7.40	5.72	17.49	13.02
0.85 mm	15.3				24.98	17.57
0.60 mm	11.3		11.73	8.64	18.24	14.30
0.30 mm	13.5		18.99	7.29	16.23	16.80
0.212 mm	5.63					
0.106 mm	7.00				7.48	11.33
Pan	6.28				2.73	2.93
Mean particle size, mm	1.24		1.77	3.05	1.24	1.31
Mean particle size, SD	0.48		0.39	0.39	0.40	0.40

¹aNDFom = NDF on an ash-free basis; TSI = total sugars as invert; WSC = water-soluble carbohydrates; unaccounted = 100% of DM minus the sum of CP, aNDFom, starch, WSC+TSI, ether extract, and ash. In addition to other uncharacterized components and analytical error, the “unaccounted” fraction may contain pectic substances, galactans, organic acids, and so on.

and >2), cows were randomly assigned to 1 of 6 dietary treatments. Feed offered and orts were recorded daily throughout the experiment.

Milk weights were recorded for all milkings during collection days, and samples for milk composition analysis were collected from all milkings on d 2 through 4 of each sampling week. Production of ECM was calculated by the following equation (derived from Tyrrell and Reid, 1965, using an energy value of 700.6 kcal/kg of milk):

$$\text{ECM, kg/d} = (12.82 \times \text{fat, kg/d}) + (7.13 \times \text{protein, kg/d}) + (0.323 \times \text{milk, kg/d}).$$

This is the same equation used to calculate ECM in the NDS Professional diet formulation model. Daily yield of NE_L (Mcal) in milk was calculated as follows: milk, kg $\times$ (0.0929 $\times$ fat % + 0.0563 $\times$ true protein % + 0.0395 $\times$ lactose %) (NRC, 2001). Feed efficiency was calculated as ECM divided by DMI. Efficiency of dietary N utilization was calculated as (milk true protein, kg/6.38)/(DMI, kg $\times$ dietary CP %/6.25).

Body condition scoring was performed as described in the Dairy NRC (2001).

Sample Collection and Analysis

Procedures and analyses are shown in Table 3, with additional details and explanation provided here. The Whatman paper and glass fiber filters listed are products of Cytiva Life Sciences. Daily DMI was calculated for each animal on a 105°C DM basis as the difference between weights of feed offered and orts. Subsamples of silages were obtained daily, frozen, and composited for each diet during the 7-d collection phase of the covariate period and experimental wk 8, and concentrate samples were collected once during those weeks. Compositional analyses were performed on feeds dried at 55°C. Chemical compositions of TMR were calculated based on the proportions of individual feeds in the diets and their chemical analyses. The compositional and digestibility analyses used in the nutritional model evaluations were performed by Dairyland Laboratories Inc. (Arcadia, WI).

Fecal samples for use in apparent total-tract digestibility analysis were obtained on d 1 through 3 of each sampling period. A total of 6 samples per cow per sampling week were obtained upon defecation or manually, with 2 samples taken per day. Sampling on d 2 and 3 were offset by 4 h from the day before to give 6 samples

Table 3. Methods and procedures used¹

Analyte or procedure ¹	Samples ²	Modifications and details	Citation
DM DM	Fd, Fc, O Mol	105°C forced-air Official Method 966.20; Karl Fischer method	Hall, 2013 AOAC, 1996 ³
Particle size	Fd	Vertical dry sieving; see notes in Materials and Methods	Hall, 2013; Waldo et al., 1971
ADF (modified)	Fd, Fc	Official Method 973.18; Modification: sand filtration aid	AOAC, 1996
NDF	Fd, Fc, O	Ash-free; heat-stable α -amylase and sodium sulfite used; sand filtration aid; ashed at 550°C for 2 h	Mertens, 2002
Sulfuric acid lignin (modified)	Fd	Official Method 973.18; Modification: sand and Whatman (Cytiva) GF/D glass fiber filter filtration aids	AOAC, 1996
NDF digestibility, undigested NDF	Fd	Fermented in vitro; strained rumen fluid and blended ruminal solids inoculum; analyzed by NDF procedure; Whatman GF/C glass fiber filters	Goering and Van Soest, 1970; Inoculum: Hall and Mertens, 2008
Ash	Fd, Fc, Mol	Official Method 942.05; 600°C for 2 h.	AOAC, 1996
WSC	Fd	Sucrose used as the carbohydrate standard	Extract: Deriaz, 1961; Detection: Hall, 2014
TSI	Mol	Official Method 968.28; Lane-Eynon method	AOAC, 1996
Starch (modified)	Fd	Modification: used YSI 2700 Select Biochemistry analyzer (YSI Inc.) for glucose detection	Hall, 2015
Starch digestibility	Fd	Fermented in vitro as for NDF digestibility; 4-mm sample grind in a cutting mill; analyzed by starch procedure	Richards et al., 1995
CP, NDIN, and ADIN expressed as N $\times$ 6.25	Fd	Official Method 990.03; Dumas combustion analysis; NDF and ADF residues collected on Whatman 541 filter paper or Whatman GF/C glass fiber filter	AOAC, 1996
Milk composition and SCC	M	Official methods 972.16 and 978.26; Infrared analysis; Foss FT6000 instrument (Foss Electric A/S); performed by AgSource Milk Analysis Laboratory (Menomonie, WI)	AOAC, 1996
Ether extract	Fd	Official Method 920.39; used diethyl ether	AOAC, 1996
Minerals	Fd	Official Method 953.01 by inductively coupled plasma emissions; Cl^- with a Corning 926 chloride analyzer (Corning Scientific; Ag^+ titration)	AOAC, 1996; Mullins et al., 2007
Apparent total-tract digestibility of DM, OM, and NDF	TMR, Fc	Ruminal in situ; internal marker: residual ADF with ash after 288 h incubation in F57 filter bags (Ankom Technology); samples dried at 55°C in a forced-air oven before subsequent analysis	Broderick and Radloff, 2004
Sample collection, processing		During collection weeks of covariate and experimental periods	Hall, 2013
DMI	Fd, O	Feed offered – orts; 105°C dry basis	Hall, 2013
Behavior	Cow	Rumination, eating, standing, laying; 24 h every 5 min	Hall, 2013

¹Unless indicated, procedures were not modified from the cited protocol; modification details are here and in the Materials and Methods section. TSI = total sugars as invert; WSC = water-soluble carbohydrates.

²Samples: Fc = feces; Fd = feed; M = milk; Mol = molasses; O = orts.

³AOAC International, 2006.

representing every 4 h of the day. Samples were dried at 55°C in a forced-air oven and ground to pass the 1-mm screen of a Wiley mill (Arthur H. Thomas Co.).

Particle sizes of forages and starch sources were calculated according to the procedure of Waldo et al. (1971). Cumulative percentages are expressed as the normal inverse on the y-axis and $\log_{10}$ of the screen size (μm) on the x-axis. In Excel the function command to calculate the normal inverse is NORMINV(probability, mean, SD), with probability = cumulative percent on a screen, mean = 0, and SD = 1. Cumulative percentages on screens of <2% or >98% were omitted from calculations (J. L. Firkins, The Ohio State University; personal communication).

Hypothetical estimated production of VFA at 3 and 9 h after feeding from Mol and CG in +RDP diets was calculated as follows: Estimated amount of feed consumed $\times$ concentration of the analyte (starch or sugar) provided by the feed of interest (CG or Mol) $\times$ proportion of the substrate calculated to be digested in a given number of hours (which gives OM fermented ruminally) $\times$ average yield of 8.03 VFA mol/kg ruminally fermented OM (Nozière et al., 2011).

Intake estimates used for 3- and 9-h time frames post-feeding were the least squares means DM intakes of cows in this study multiplied by 30% and 60%, respectively, which were the average proportions of daily ration consumed at those times measured on lactating Holstein cows in a separate study on a diet that did not contain molasses (M. B. Hall, unpublished data). The primarily slower-fermenting feeds not used to create treatments (corn silage, alfalfa silage, distillers grains, soyhulls, and soybean meal) accounted for total daily DMI differences of 2.20 and 1.05 kg of DM for +RDP diets with 10.5 and 5.25% Mol versus +RDP with 0% Mol. At 3 through 9 h after feeding, at 30 to 60% of ration consumption, they would have accounted for, at most, 0.66 to 1.32 kg of DM. These feeds were not used in the calculations of VFA produced in early hours after feeding. From the NDS model, fermentation rates (kd) were 8.45%/h for starch in the corn grain and 60%/h for sugar in molasses; passage rates (kp) used for those fractions were 4% and 10%/h, respectively, as approximate kp derived from the literature (e.g., Yang et al., 2002). The combined rate of sugar or starch fermented in or passing from the rumen equals (kd + kp). The proportion of substrate that was fermented in any time frame equals $\text{kd}/(\text{kd} + \text{kp})$. The amount of substrate fermented in a given time frame is calculated using the equation $[\text{kd}/(\text{kd} + \text{kp})] \times [D_0 - (D_0 \times e^{-kt})]$, where D_0 = amount of substrate consumed by the cow in a given time frame, $k = (\text{kd} + \text{kp})$, and $t = 3\text{- or }9\text{-h time frame}$. Statistics were not applied to these estimates.

The Dairy NRC model (NRC, 2001) and NDS Professional/CNCPS 6.55 (CNCPS/NDS) were used to generate nutrient supply and ECM predictions for the diets and were compared with actual animal performance. The ECM predictions were based on MP, ME, or NE_L . Data used in the models by dietary treatment were arithmetic averages for actual DMI, diet composition, analyses of the individual feeds, and cow characteristics. The NRC allowable milk values were converted to ECM as described previously. Because 8.7% of DM in molasses does not analyze in the proximate analysis system and is likely to be the indigestible products of the alkali and heat processing of sugarcane (Binkley and Wolfrom, 1953) that are nonnutritive to the microbes or the cow, we modified entry of its composition in the models so that it fell into nonnutritive fractions in each model. The 8.7% of Mol DM was added to the ash in the NRC model and was treated as indigestible starch (rate of fermentation of 0.001% and 0% small intestinal digestibility) in CNCPS/NDS. In this way, it would not incorrectly inflate the predicted nutrient supply as NFC by difference in NRC or as soluble fiber by difference in CNCPS/NDS, which would bias calculated nutrient contributions in diets as Mol levels increased.

Statistical Analysis

Variables used in the statistical model were Y = dependent variable, μ = overall mean, M_i = dietary level of molasses ($i = 0, 5.25, 10.5$), MM_i = quadratic term of M_i , R_j = RDP treatment ($j = \text{soybean meal: +RDP; expeller soybean meal: -RDP}$), MR_{ij} = interaction term for M_i and R_j , MMR_{ij} = interaction term for quadratic term of M_i and R_j , C_k = covariate period value for the dependent variable, DIM_i = DIM at the start of experimental feeding wk 8, L_m = lactation group by parity ($m = 1, 2, \geq 3$), and ε = residual error. Continuous variables included M_i , C_k , and DIM_i ; the remaining variables were classification variables. No variables were treated as random variables. The differences between actual and nutritional model predictions of ECM were analyzed with the model, where Y = dependent variable, μ = overall mean, M_i = dietary level of molasses, MM_i = quadratic term of M_i , R_j = RDP treatment, MR_{ij} = interaction term for M_i and R_j , MMR_{ij} = interaction term for quadratic term of M_i and R_j , N_m = nutritional model ($m = \text{NRC or CNCPS/NDS}$), MN_{im} = interaction of M_i and N_m , RN_{jm} = interaction of R_j and N_m , and MRN_{ijm} = interaction of M_i , R_j , and N_m . Models were reduced when the significance of a term was $P > 0.25$, in which case the term was removed; M_i , R_j , their interaction, N_m , and C_k were

always retained. Results are reported as least squares or arithmetic means as indicated. Least squares means and standard errors of the difference were determined in models containing only the linear terms for Mol used as a classification variable. Significance was declared at $P \leq 0.05$ and tendency at $0.05 \leq P \leq 0.10$ for main effects, and tendency for interaction terms was extended to $0.05 \leq P \leq 0.15$. Analyses were performed with the MIXED procedure of SAS (Version 9.4, 2016, SAS Institute Inc.). Normality of the residuals was evaluated with the Shapiro-Wilk test at $P < 0.05$ using PROC UNIVARIATE of SAS; data were transformed to normalize. Residuals of lactose percentage data could not be made normal through transformation; a quartic transformation appeared to make all points except 1 extreme observation fall on the normal probability plot reference line.

RESULTS

Intake and Lactation Performance

Dry matter and OM intakes declined linearly ($P \leq 0.04$) with increasing Mol and decreasing CG content of diets, with no detected carbohydrate source $\times$ RDP interaction ($P \geq 0.37$), nor an effect of RDP treatment ($P \geq 0.73$; Table 4).

Milk and ECM production and NE_L in milk declined linearly with increasing Mol ($P \leq 0.05$), but we did not detect effects of RDP ($P \geq 0.63$) or of a carbohydrate source $\times$ RDP interaction ($P \geq 0.36$) on these measures. Yields of protein, lactose, and SNF declined linearly with increasing Mol ($P \leq 0.02$). Percentages of both protein and SNF showed similar patterns for the quadratic interaction of Mol and RDP ($P \leq 0.01$): relative to 0% Mol, with +RDP we observed a decline at 5.25% Mol and then a rise at 10.5% Mol, and with -RDP we found little change or slight decline between 0 and 5.25% Mol and then decline at 10.5% Mol. Although the differences in protein and SNF percentages were significant, they were numerically small. Lactose percentage tended to show a linear interaction of Mol $\times$ RDP: it changed little across Mol levels with +RDP but declined with increasing Mol with -RDP ($P = 0.11$). Milk urea N was unaffected by treatment ($P \geq 0.14$). The only detected effect of RDP alone was for lower values of SCC ($P < 0.01$) with +RDP diets; SCC also showed a tendency for a linear treatment interaction effect ($P = 0.11$). In contrast to the other milk components, milk fat yield did not differ among treatments ($P \geq 0.23$). To achieve these fat yields given declining milk production, milk fat percentage increased linearly with increasing Mol ($P < 0.01$), with a tendency for a linear interaction of Mol $\times$ RDP ($P = 0.08$); as Mol

increased, milk fat percentage rose with +RDP but changed relatively little with -RDP.

The efficiency terms of milk NE_L or ECM divided by DMI or OM intake did not differ among treatments ($P \geq 0.25$; Table 4). We detected a linear decline in milk nitrogen efficiency ($P = 0.04$) with increasing Mol. Cows did not display detectable differences among treatments in BCS or BW changes ($P \geq 0.33$).

Apparent Diet Digestibility

Total-tract apparent digestibilities of DM, OM, and NDF were all affected by Mol (Table 5). The Mol treatments showed quadratic patterns for all digestibilities ($P \leq 0.05$), with the lowest value always at the Mol 5.25% treatment. Mass of OM digested declined linearly with increasing Mol intake ($P = 0.03$). This coincided with declining DMI (Table 4), increasing ash contents of the diets (Table 1), and linear increases in fecal ash content with increasing Mol inclusion ($P < 0.01$). The efficiency of NE_L Mcal in milk per kilogram of OM apparently digested did not differ among treatments ($P \geq 0.40$). The effects of RDP alone ($P \geq 0.60$) or of Mol $\times$ RDP ($P \geq 0.19$) were not significant for any comparison.

Model Predictions

All factors tested in the statistical model evaluating the ECM differences between actual and model predictions based on MP were significant (Table 6, Table 7). Compared with the actual performance of cows in the study, the NRC and CNCPS/NDS model predictions based on MP generally underestimated allowable ECM production for +RDP diets and overestimated or were similar for -RDP diets (RDP, $P < 0.01$; Table 6, Figure 1; values presented here and in tables are arithmetic averages or differences). Based on predicted MP, on average the NRC model underpredicted ECM on +RDP diets by 4.5 kg, and CNCPS/NDS underpredicted by 2.3 kg, whereas with -RDP diets, both models overestimated ECM, with CNCPS/NDS overestimating by 5.0 kg and NRC coming close to actual performance values at +0.25 kg (RDP $\times$ model, $P < 0.01$). The Mol $\times$ RDP $\times$ model interaction ($P < 0.01$) for the difference between actual and model-predicted ECM based on MP reflects differing response patterns to Mol levels within RDP treatment by the nutritional models (Figure 1). The responses to Mol appear to be quadratic (Mol² $\times$ RDP, $P < 0.01$), but there were insufficient degrees of freedom to test the interaction that included RDP, nutritional model, and the quadratic term for Mol.

Table 4. Cow performance, with values as least squares or arithmetic means

Measure ¹	+RDP ²			-RDP ²			P-value ⁴			
	Molasses, % of diet DM			Molasses, % of diet DM			Mol		Mol × RDP	
	0	5.25	10.5	0	5.25	10.5	RDP	L	Q	L
DMI, kg	29.0	27.9	26.7	28.8	27.8	27.9	1.09	0.04	0.79	0.37
OMI, kg	26.6	25.5	24.4	26.5	25.5	25.5	1.00	0.03	0.79	0.38
Milk, kg	47.7	44.8	42.3	48.3	44.7	44.2	1.81	<0.01	0.44	0.60
Fat, kg	1.59	1.68	1.65	1.68	1.58	1.57	0.104	0.74	0.94	0.23
Fat, %	3.39	3.75	3.93	3.48	3.47	3.59	0.172	<0.01	0.91	0.08
Protein, kg	1.51	1.38	1.36	1.50	1.38	1.32	0.053	0.01	0.18	0.71
Protein, %	3.18	3.08	3.19	3.10	3.10	2.98	0.053	0.31	0.53	0.04
Lactose, kg	2.36	2.20	2.15	2.45	2.18	2.13	1.48	0.02	0.19	0.75
Lactose, %	4.94	4.95	5.03	4.97	4.95	4.87	20.8	0.13	0.35	0.11
SNF, kg	4.26	3.95	3.80	4.29	3.95	3.85	0.162	<0.01	0.33	0.86
SNF, %	8.96	8.83	8.97	8.90	8.87	8.71	0.058	0.10	0.29	0.05
SCC	44	255	247	92	323	209	0.015	0.12	0.32	0.11
MUN, mg/dL	12.4	12.2	12.3	12.2	11.7	12.0	0.005	0.14	0.20	0.78
ECM, kg	46.5	45.8	44.7	47.9	44.3	43.8	2.08	0.05	0.62	0.42
ECM/DMI	1.63	1.66	1.66	1.67	1.62	1.60	0.073	0.70	0.95	0.31
ECM/OMI	1.77	1.81	1.82	1.82	1.77	1.75	0.080	0.82	0.95	0.31
Milk NE _L , Mcal	32.6	32.0	31.4	33.6	31.0	30.5	1.48	0.05	0.58	0.36
Milk NE _L /DMI	1.14	1.16	1.17	1.17	1.13	1.12	0.052	0.68	0.92	0.25
Milk NE _L /OMI	1.24	1.27	1.28	1.28	1.24	1.22	0.056	0.79	0.92	0.25
MN/IN	0.310	0.292	0.297	0.308	0.297	0.285	0.012	0.04	0.49	0.52
BCS change	-0.05	-0.08	0.00	-0.02	-0.00	-0.10	2.87	0.65	0.50	0.40
BW change, kg	22.4	18.5	17.1	20.3	24.6	13.1	9.0	0.33	0.55	0.89

¹OMI = OM intake; IN = intake nitrogen; MN = milk nitrogen.²RDP treatments: +, with more RDP; -, with less RDP.³SED = standard error of the difference; values for transformed data are not back-transformed.⁴Effects of molasses (Mol): L = linear; Q = quadratic. Data transformations used were SCC and MUN as 1/√(Y), lactose (kg) as Y³. BCS change was ultimately analyzed as 1/Y and lactose percentage as Y⁴; each had one heavy tail that could not be transformed to be made normal. Values shown are arithmetic means.

Table 5. Total-tract apparent digestibilities, apparently digested OM, fecal ash concentration, and milk NE_L/digested OM

Measure ¹	+RDP ²			−RDP ²			SED ³	RDP	P-value ⁴			
	Molasses, % of diet DM			Molasses, % of diet DM					Mol		Mol × RDP	
	0	5.25	10.5	0	5.25	10.5			L	Q	L	Q
DM, % of DM	72.4	72.0	73.5	72.4	70.8	74.7	1.11	0.95	0.05	<0.01	0.30	0.19
OM, % of OM	74.2	73.5	75.0	74.0	72.4	75.8	1.07	0.81	0.03	<0.01	0.43	0.30
NDF, % of NDF	52.8	50.4	51.8	52.4	50.4	54.6	1.99	0.86	0.08	0.05	0.87	0.63
OMD, kg/d	20.0	18.6	18.5	20.0	18.5	19.1	0.81	0.83	0.03	0.09	0.60	0.67
Ash, % of fecal DM	12.2	12.9	13.0	12.0	12.8	13.1	0.29	0.61	<0.01	0.09	0.84	0.69
Milk NE _L /OMD kg	1.66	1.73	1.69	1.70	1.70	1.63	0.09	0.60	0.47	0.40	0.68	0.86

¹OMD = apparently digested OM; Milk NE_L = Mcal in milk produced.²RDP treatments: +, with more RDP; -, with less RDP.³SED = standard error of the difference.⁴Mol and Mol × RDP treatment responses: L = linear, Q = quadratic effects of Mol.

For predicted energy allowable ECM production, the models differed overall in their deviations from actual ECM (1.2 and 3.0 kg for NRC and CNCPS/NDS, respectively; $P < 0.01$). Model effects differed by Mol (Mol × model, $P = 0.04$; deviation ECM kg for NRC: 1.1, 1.6, and 1.1, and for CNCPS/NDS: 3.6, 3.2, and 2.3 for 0, 5.25, and 10.5% Mol, respectively), and by

RDP (RDP × model, $P = 0.05$; deviation ECM kg for NRC: 0.2 and 2.3, and for CNCPS/NDS: 1.7 and 4.4 for +RDP and -RDP, respectively). We detected a 3-way interaction of Mol × RDP × model ($P = 0.07$). Compared with MP-based predictions, ECM predictions based on energy were numerically closer to actual ECM with +RDP diets. The NRC model flagged NE_L

Table 6. Actual cow performance (as arithmetic means) and model predictions

Measure	+RDP ¹			-RDP ¹		
	Molasses, % of diet DM			Molasses, % of diet DM		
	0	5.25	10.5	0	5.25	10.5
Actual performance ²						
DMI, kg	28.1	28.1	26.5	30.0	27.7	27.9
Milk, kg	47.8	44.5	42.7	49.2	44.2	43.6
Milk fat, %	3.31	3.75	3.85	3.51	3.62	3.52
True protein, %	3.13	3.09	3.20	3.07	3.05	3.03
ECM, kg	46.3	45.6	44.6	48.8	44.4	43.2
CNCPS/NDS predictions ³						
ME, Mcal/d	73.5	73.2	68.8	78.5	72.3	72.3
MP, g/d	2,986	2,993	2,812	3,451	3,173	3,210
ME allowable ECM, kg	48.8	48.1	44.8	53.6	48.2	47.6
MP allowable ECM, kg	42.4	45.2	42.1	53.0	49.3	49.1
MP from bacteria, g	1,533	1,573	1,532	1,595	1,528	1,570
MP from RUP, g	1,452	1,420	1,280	1,856	1,645	1,640
Rumen balance: NH ₃ N g	150	145	134	112	98	93
Rumen balance: Peptide N g	262	258	244	224	206	202
NRC predictions ⁴						
NE _L , Mcal/d	44.0	43.8	41.5	46.3	43.3	43.2
MP, g/d	2,918	2,911	2,741	3,236	2,993	3,034
NE _L allowable ECM, kg	47.0	46.7	43.4	50.3	46.4	46.5
MP allowable ECM, kg	40.4	42.9	39.7	47.4	44.6	45.1
MP from bacteria, g	1,499	1,494	1,414	1,492	1,401	1,420
MP from RUP, g	1,286	1,284	1,201	1,603	1,462	1,483
RDP balance, g/d	20	56	89	-169	-139	-102
RUP balance, g/d	-433	-177	-335	-88	15	120

¹RDP treatments: +, with more RDP; -, with less RDP.²Actual performance values are arithmetic means of the study data.³CNCPS/NDS = NDS (Nutritional Dynamic System) Professional/CNCPS (Cornell Net Carbohydrate and Protein System) 6.55, RUM&N Sas.⁴NRC, 2001.

Table 7. *P*-values from the full statistical models for the evaluation of the difference between actual cow performance and model predictions based on energy- or metabolizable protein-based predictions

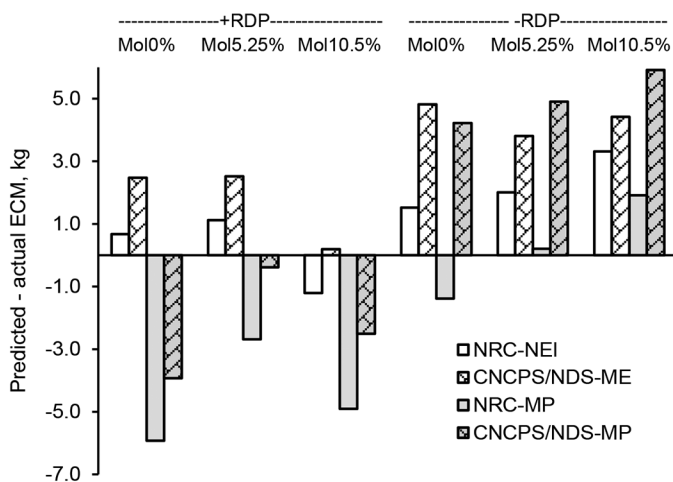
Variable ¹	ME	MP
Mol	0.29	<0.01
Mol × Mol	0.09	<0.01
RDP	0.01	<0.01
Mol × RDP	0.04	<0.01
Mol × Mol × RDP	0.01	<0.01
Model	<0.01	<0.01
Mol × Mod	0.04	0.02
RDP × Mod	0.05	<0.01
Mol × RDP × Mod	0.07	<0.01

¹Mol = molasses level substituted for corn grain treatment; Mod = nutritional model, NRC (2001) or CNCPS/NDS [NDS (Nutritional Dynamic System) Professional/CNCPS (Cornell Net Carbohydrate and Protein System) 6.55, RUM&N Sas]. Interaction terms include quadratic effects of Mol.

allowable ECM predictions for –RDP diets as possibly being incorrect because that model predicted that RDP was deficient in those diets.

Behavior

Treatment effects on cow behavior were few (Table 8). Rumination minutes per day tended to show a relatively flat response over +RDP Mol treatments but declined with increasing Mol on –RDP (linear Mol × RDP, *P* = 0.11). The Mol treatments showed a linear increase in the number of episodes of eating per day (*P* = 0.01) despite a concomitant linear decrease in DMI (*P* = 0.04). A tendency for the effect of the lin-

**Figure 1.** Model-predicted minus actual ECM. Mol = molasses treatment, at 0, 5.25, or 10.5% of diet DM. RDP treatments: +, with more RDP; –, with less RDP. NRC = predicted from NRC, 2001; CNCPS/NDS = predicted from NDS (Nutritional Dynamic System) Professional/CNCPS (Cornell Net Carbohydrate and Protein System) 6.55, RUM&N Sas.**Table 8.** Behavior measures

Item	+RDP ¹				–RDP ¹				<i>P</i> -value ³			
	Molasses, % of diet DM				Molasses, % of diet DM				Mol			
	0	5.25	10.5		0	5.25	10.5		L	Q	L	Q
Minutes per day												
Lying	698	717	673		670	685	731		0.61	0.80	0.25	0.44
Ruminating	392	414	394		438	390	375		0.76	0.89	0.11	0.25
Eating	185	179	172		193	181	209		0.84	0.34	0.22	0.34
Episodes per day												
Lying	8.0	7.5	8.3		9.2	8.8	9.1		0.96	0.39	0.80	0.80
Ruminating	10.9	12.3	11.3		11.5	11.2	11.0		0.26	0.23	0.15	0.21
Eating	5.5	6.9	7.0		5.9	6.5	7.3		0.01	0.71	0.96	0.42
Minutes per kg of DMI												
Ruminating	14.0	15.2	14.4		15.0	14.3	13.8		0.71	0.62	0.38	0.47
Eating	6.54	6.56	6.44		6.84	6.72	7.84		0.34	0.53	0.25	0.40

¹RDP treatments: +, with more RDP; –, with less RDP.

²SED = standard error of the difference.

³Mol and Mol × RDP treatment responses: L = linear, Q = quadratic effects of Mol.

ear interaction of Mol $\times$ RDP ($P = 0.15$) on episodes of rumination per day showed a relative increase as Mol increased with +RDP, but a small decrease with -RDP. No differences were noted among treatments for minutes spent ruminating or eating per kilogram of DMI ($P \geq 0.25$).

DISCUSSION

Corn Grain and Molasses Comparability

Corn grain and cane molasses differ greatly in nutrient composition and digestion characteristics, making it informative to evaluate cow responses to these feeds within the context of their differences (Table 2). In addition to obvious dissimilarities of having dry versus liquid forms, and of being predominantly starch versus sugar sources, they differ substantially in nutrient content. Molasses contains approximately 14 times as much ash as corn grain and can be a substantial source of potassium. Up to 10% of DM in molasses that is unaccounted for by nutrient analyses may consist of compounds including Maillard products that have little or no direct nutritional value and are formed during sugar cane processing by reactions of carbohydrates under heat and alkaline pH, and with AA (Binkley and Wolfrom, 1953). The by-difference inclusion of these materials with the most fermentable carbohydrates, in NFC in the NRC model and in soluble fiber in CNCPS/NDS, is a flaw in the correct allocation of materials in these models. Our decision to apportion the materials to indigestible fractions in the models was made in an effort to allow the models to more accurately calculate carbohydrate fractions and their nutritive value to the cow. To have done otherwise would simply perpetuate a recognized error. In terms of AA, blackstrap cane molasses has been reported to contain approximately 4% of DM as CP and 2.3% of DM as free AA measured after acid hydrolysis (Mee et al., 1979).

The actual difference in carbohydrate content between cane molasses and corn grain is greater than it first appears. The 60.4% of DM as total sugars as invert in molasses is on a monosaccharide basis, whereas the 71.9% starch in corn grain is in a polysaccharide form. The starch content is converted to a free monosaccharide basis by dividing the starch value by 0.9, which adds back the weight of water needed for hydrolysis of each bond between anhydro-glucose monomers. This gives corn grain a monosaccharide content from starch of 79.9% of DM. Combining starch monosaccharides plus 3.55% of DM from water-soluble carbohydrates gives corn grain 38% more potentially digestible carbohydrate than molasses, not counting available NDF.

However, with no physical or insolubility barriers, the ruminal rate of NFC utilization is likely much more rapid for sugars in molasses than for starch in corn grain. It has been reported that sugar in substrate containing added sugar was more digestible in continuous culture than the sugars present in the plant materials used (Hoover et al., 2006). The rate of starch fermentation in corn grain is affected by particle size (Galyean et al., 1981), the protein matrix associated with the starch granules (Philippeau et al., 2000), and the water-excluding nature of the native starch granules (Pérez et al., 2009). Given that no other nutrients are limiting, the differences in rates of starch and sugar fermentation could result in a dilution of maintenance effect for the ruminal microbes utilizing the more rapidly fermenting free sugars, as a greater amount of substrate would be available in excess of maintenance per unit of time. Similarly, it could also create differences in the amounts of VFA available to the cow in different time frames.

Lactation Performance

The decline in intake with increasing Mol inclusion in the present study was not expected. Many studies have reported increases in intake associated with Mol or sugar replacing CG or starch (Yan et al., 1997; Broderick and Radloff, 2004; Broderick et al., 2008). Decline in intake with increasing Mol and decreasing CG had been noted by Ghedini et al. (2018) feeding 0 to 12% liquid molasses in diets containing 52% baleage and low in starch (1.5 to 9.5% of DM). Broderick and Radloff (2004) reported an initial increase in DMI with addition of 3% liquid molasses, but subsequent decline as molasses was increased to 6 and 9% of DM on diets of 32% alfalfa silage, 20% corn silage, and with 26 to 29% of diet DM as starch. Although some studies have reported a DMI response to increasing RDP (Herrera-Saldana and Huber, 1989; Hall, 2013), +RDP in the current study did not have that effect. No factors measured in the present study provided a basis to explain the molasses-related decline.

The declines in milk and some component yields appear to be driven primarily by declining DMI, given that the ECM and milk NE_L production efficiency terms did not differ among treatments. Given the lack of detected differences among treatments for changes in BCS or BW, the similarity in feed efficiencies across treatments suggests that the diets provided energetically similar quantities of nutrients derived per unit of DM. However, it seems clear that the identity of those nutrients and their utilization were not the same across diets, as evidenced by the maintenance of milk fat yield and decreases in protein and lactose yields with

increasing molasses and decreasing DMI. The linear decline in milk nitrogen efficiency and increase in milk fat percentage with increasing Mol is consistent with the decline in nitrogen efficiency and increase in milk fat noted when sucrose replaced corn starch in diets (G. A. Broderick, USDA-ARS; personal communication; Broderick et al., 2008). It appears that milk fat versus milk protein and lactose production responses moved separately with substitution of molasses for corn grain.

A partial explanation for these results may be related to cow-microbe-diet dynamics over time. Although we typically describe cow responses to diets on a “day” basis, normally fed and managed dairy cows are not in steady state, nor is “day” the appropriate time step for all of their digestion or metabolic processes. In contrast, we often track ruminal responses on an hourly basis. A shorter time step describing the interaction of the cow, rumen microbes, and diet is the basis for a hypothesis to explain the milk fat and intake responses in the present study.

The VFA acetate and butyrate are crucial to milk fat production. Acetate is important for elongation of de novo synthesized milk fat FA, and butyrate provides approximately 50% of initial 4-carbon units in de novo synthesized FA (Palmquist et al., 1969) and 30% of the FA in the *sn*-3 position on milk triglycerides (Jensen, 2002). Ruminal fermentation of sugars gives a greater molar proportion of butyrate than do other carbohydrate sources. It may be this and the combination of (1) rate of ruminal influx of substrate and (2) rate and mass of VFA production from the rapid fermentation of sugars that create a temporal excess of VFA relative to the existing (“basal”) energetic and synthetic needs of the lactating cow. With cells being incapable of storing ATP, under these conditions we hypothesize that fat production may be increased by shunting the acetate and butyrate that is in excess of basal needs to de novo FA synthesis.

In the early hours after feeding, the most rapidly fermentable NFC likely represent the most significant sources of VFA. Our hypothetical calculated estimates of ruminal VFA production from Mol + CG on the RDP diets, based on DM intakes from the present study, the proportion of daily intakes consumed at 3 and 9 h after feeding (measured in another study), and rates of fermentation for sugar from Mol and starch from CG give VFA moles produced by 3 h post-feeding of 1.9, 3.0, and 4.0 for 0, 5.25, and 10.5% Mol diets, respectively. The 10.5% Mol treatment gave approximately double the estimated VFA produced from the 0% Mol treatment. By 9 h, the calculated VFA moles produced from Mol and CG were 8.4, 9.8, and 11.1 for 0, 5.25, and 10.5% Mol diets, respectively. If these estimated increases in VFA produced were in excess of

the basal energetic and synthetic needs of the cow in that time frame, it is possible that the excess acetate and butyrate were shunted to fat synthesis. The animal would need to have the synthetic capacity to convert VFA to FA, and other factors such as physiologic state would likely affect whether VFA were diverted to milk fat or to adipose. Milk fat production and de novo FA synthesis have been shown to be increased with ruminally infused acetate (Urrutia and Harvatine, 2017) and butyrate (milk fat production: Huhtanen et al., 1993; de novo FA synthesis: Rico et al., 2021). Shunting of acetate and butyrate to milk fat would also explain why milk fat production diverged from that of milk protein and lactose in the present study: these VFA are not used for production of those milk components. Lastly, the shunting of VFA to milk fat rather than to energy production could have played a role in maintaining the ECM and NE_L efficiency metrics, because there would have been no loss as heat increment. The temporal effects of VFA supply on animal performance need to be explored further.

For the reduced DMI noted as Mol inclusion increased, we offer a hypothesis similar to that described for the milk fat response. The peak in VFA mass delivered to the cow in early hours after feeding includes propionate, which has been shown to depress DM intake (Shepherd and Combs, 1998). Production of propionate that temporally exceeds a threshold at which intake is depressed could explain the reduced intakes on the present study. Molar percentages of VFA from sucrose fermented in vitro with inoculum from lactating Holstein cows averaged 42%, 42%, and 15% for acetate, propionate, and butyrate, respectively (values do not sum to 100 due to rounding; M. B. Hall, unpublished data). Sucrose can ferment to lactate (Strobel and Russell, 1986), and lactate can be converted by ruminal microbes to propionate and acetate, but preferentially to propionate in diets containing high levels of starch-containing concentrates or lactate (Baldwin et al., 1962).

Potential bases for the differing results in DMI and milk fat responses of this study as compared with the companion ruminal study (Zanton and Hall, 2022) could be related to DIM and to feeding protocols. The cows in the present study averaged 101 DIM as compared with 185 for the other study. It is possible that cows on the present study had greater synthetic capacity to produce milk fat and were at a physiologic state that allowed preferential diversion to milk fat rather than adipose. The feeding management in the current study was such that the cows were fed their entire ration at 1 time per day and the feed was pushed up to them through the day. In the companion study (Zanton and Hall, 2022), cows were fed 50% of their daily ration twice daily at 12-h intervals, with most of the feed from the first feed-

ing having been consumed by the time of the second feeding. If the hypothesis regarding the influence of the timing of ruminal VFA production based on the rate of animal feed consumption and composition is correct, the twice-a-day feeding would have restricted the cows' intake to no more than 50% of the ration in the first 12 h after initial feeding, which could have affected moles of peak VFA production.

Another effect of sugars that may be invoked related to milk fat production is ruminal biohydrogenation. Sugar (glucose, maltose, cellobiose)-utilizing microbes in the rumen are capable of biohydrogenation of FA (McKain et al., 2010) that can reduce the amount of bioactive unsaturated FA reaching the cow that could depress milk fat production (Shingfield and Griinari, 2007). However, little fat was included in the diets of the present study (3.7 to 4.5% of DM), making the influence of biohydrogenation unlikely to be a primary factor in the fat responses.

Model Predictions

It is of concern that both the NRC (2001) and the CNCPS/NDS models underestimated ECM on the diets formulated to be higher in RDP when using an MP allowable ECM basis (Figure 1). This suggests that MP supply, either as microbial protein or as escape feed protein, is being systematically underestimated by both models when more RDP is included in the diet, essentially underestimating the value of RDP to high-producing cows. The relative changes in predicted MP sources showed different patterns and magnitudes between the models and, not surprisingly, appeared to be affected by DM intake (not evaluated statistically; Table 6). The model outputs for predicted microbial MP show an average 31-g decline between +RDP and -RDP diets for NRC, and an average 18-g increase with increased RUP for CNCPS/NDS; the average changes between +RDP and -RDP diets in MP from RUP were +259 g for NRC and +330 g for CNCPS/NDS (Table 6). Assuming that MP requirements for ECM are correct, it cannot be discerned from the information we have whether underestimated predictions of available MP from RUP or microbial protein or both are the issue for +RDP predictions. It was interesting that, of the 2 models, only the NRC model flagged the -RDP diets for being inadequate in RDP. However, based on other measurements, the NRC model prediction of deficiency may have been incorrect. The MUN values in the present study averaged 12.1 mg/dL. This is comparable to the values noted in the companion study (Zanton and Hall, 2022), in which average ruminal ammonia (5.7 mM) and branched-chain VFA (2.32 mM) concentrations did not differ among treatments.

The ammonia concentration is in a range considered to be adequate for ruminal function (Satter and Slyter, 1974).

Total-Tract Apparent Digestibility

The basis for the decline in DM, OM, and NDF digestibilities at 5.25% inclusion of molasses but not at 10.5% was not apparent. Hoover et al. (2006) similarly noted a depression in fiber digestibility with the initial addition of a sugar product at 63 g of sugar/kg of DM at 240 and 280 g of nonstructural carbohydrate/kg of DM. The negative response was not detected as sugar was increased to 95 g/kg of DM and 280 g of nonstructural carbohydrate/kg of DM in continuous culture experiments. Broderick and Radloff (2004) reported cubic responses to supplementing lactating dairy cows with 0, 3, 6, and 9% liquid molasses. The DM and OM digestibilities declined at 3% molasses, then rose at 6%, and declined slightly at 9%. In that study, NDF digestibility did not differ between 0 and 3% molasses, rose at 6%, and declined at 9%. In that same study, DM, OM, and NDF digestibilities increased linearly with increasing inclusion of dried molasses at 0, 4, 8, and 12% of diet DM. The nonlinear effect of sugar or molasses on digestibilities, with initial additions seeming to have the most marked negative effects, is perplexing, as is the reported difference in response to dried versus liquid molasses. The current research literature did not provide a ready explanation for these results.

CONCLUSIONS

The results of this study found against our hypotheses regarding effect of molasses on increasing DM intake and the agreement of the nutritional models with actual animal performance, but found for our hypothesized positive effect of molasses and RDP on milk fat percentage. The most important aspect of this study is that it raises the question of what are the bases for the intake and milk fat responses. As to the performance of the nutritional models, it suggests a re-evaluation of how the influence of estimated protein degradability affects performance in high-yielding dairy cows, particularly on an MP basis.

ACKNOWLEDGMENTS

This research was supported through a Trust Fund Cooperative Agreement with Westway Feed Products LLC (Tomball, TX). This research was also supported by funding from the USDA, Agricultural Research Service (Washington, DC) under National Program 101 Food Animal Production Current Research Informa-

tion System (CRIS) funds (project no. 5090-31000-026-00D). Mention of any trademark or proprietary product in this manuscript does not constitute a guarantee or warranty of the product by the USDA or the Agricultural Research Service and does not imply its approval to the exclusion of other products that also may be suitable. USDA is an equal opportunity provider and employer. We thank Joe Harris and Tony Tucker of Westway Feed Products LLC for their input into the design and discussions on this project, Ermanno Melli, David Weber, and the NDS Group (RUM&N Sas, Reggio Emilia, Italy) and Curt Cotanch of Barn Swallow Consulting, LLC (Underhill, VT) for their assistance in evaluating the diets in the CNCPS/NDS program, and Landus Cooperative (Ames, IA) for the donation of the SoyPlus used in this study. The conscientious work of Kurt Pickar for feeding the cows, the rest of the farm staff for caring for the cows at the Dairy Forage Research Center Farm (Prairie du Sac, WI), and Jan Pitas, Wendy Radloff, and all Dairy Forage volunteers for assistance in this study is gratefully acknowledged. The authors have not stated any conflicts of interest.

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